

## 8. String to Integer (atoi)

Medium Topics Companies

Implement the `myAtoi(string s)` function, which converts a string to a 32-bit signed integer.

The algorithm for `myAtoi(string s)` is as follows:

- Whitespace:** Ignore any leading whitespace (" ").
- Signedness:** Determine the sign by checking if the next character is '-' or '+', assuming positivity if neither present.
- Conversion:** Read the integer by skipping leading zeros until a non-digit character is encountered or the end of the string is reached. If no digits were read, then the result is 0.
- Rounding:** If the integer is out of the 32-bit signed integer range  $[-2^{31}, 2^{31} - 1]$ , then round the integer to remain in the range. Specifically, integers less than  $-2^{31}$  should be rounded to  $-2^{31}$ , and integers greater than  $2^{31} - 1$  should be rounded to  $2^{31} - 1$ .

Return the integer as the final result.

### Example 1:

Input: s = "42"

Output: 42

#### Explanation:

The underlined characters are what is read in and the caret is the current reader position.

Step 1: "42" (no characters read because there is no leading whitespace)

Step 2: "42" (no characters read because there is neither a '-' nor '+')

Step 3: "42" ("42" is read in)

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### Code

Java Auto

```
1 class Solution {
2     public int myAtoi(String s) {
3
4         int i = 0;
5         int n = s.length();
6
7         // 1. Ignore leading spaces
8         while (i < n && s.charAt(i) == ' ') {
9             i++;
10        }
11
12        // 2. Check sign
13        int sign = 1;
14
15        if (i < n && s.charAt(i) == '-') {
```

Saved

Ln 9, Col 17

Testcase Test Result

Accepted Runtime: 0 ms

Case 1 Case 2 Case 3 Case 4 Case 5

Input

s =  
"42"

Output

42

Expected

42

Contribute a testcase